

```
Code

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
Code

df=pd.read_csv("/content/Bike Sales Data.csv")
df
```

```
Output

      ID Marital Status Gender   Income  Children  Education \
0    12496.0          M      F  $40,000.00      1.0    Bachelors
1    24107.0          M      M  $30,000.00      3.0  Partial College
2    14177.0          M      M  $80,000.00      5.0  Partial College
3    24381.0          S      M  $70,000.00      0.0    Bachelors
4    25597.0          S      M  $30,000.00      0.0    Bachelors
...      ...          ...      ...      ...      ...      ...
1023  22400.0          M      M  $10,000.00      0.0  Partial College
1024  20942.0          S      F  $20,000.00      0.0    High School
1025  18484.0          S      M  $80,000.00      2.0    High School
1026      NaN          NaN    NaN      NaN      NaN      NaN
1027      0.0          NaN    NaN      NaN      NaN      NaN

      Occupation Home Owner  Cars  Commute Distance  Region  Age \
0    Skilled Manual      Yes   0.0      0-1 Miles  Europe  42.0
1      Clerical      Yes   1.0      0-1 Miles  Europe  43.0
2    Professional      No   2.0      2-5 Miles  Europe  60.0
3    Professional      Yes   1.0      5-10 Miles  Pacific  41.0
4      Clerical      No   0.0      0-1 Miles  Europe  36.0
...      ...          ...      ...      ...      ...      ...
1023      Manual      No   1.0      0-1 Miles  Pacific  26.0
1024      Manual      No   1.0      5-10 Miles  Europe  31.0
1025  Skilled Manual      No   2.0      1-2 Miles  Pacific  50.0
1026      NaN      NaN   NaN      NaN      NaN   NaN
1027      NaN      NaN   NaN      NaN      NaN   NaN

      Purchased Bike
0              No
1              No
2              No
3              Yes
4              Yes
...      ...
1023      Yes
1024      No
1025      Yes
1026      NaN
1027      NaN

[1028 rows x 13 columns]
```

```
Code

#describing the data
print("shape:",df.shape)
print("\n columns:")
print(df.columns)
print("\n Data Types:")
print (df.dtypes)
print("\n First 5 records:")
df.head()
```

```
Output
```

```
shape: (1028, 13)

columns:
Index(['ID', 'Marital Status', 'Gender', 'Income', 'Children', 'Education',
      'Occupation', 'Home Owner', 'Cars', 'Commute Distance', 'Region', 'Age',
      'Purchased Bike'],
      dtype='object')

Data Types:
ID                float64
Marital Status    object
Gender            object
Income            object
Children          float64
Education         object
Occupation        object
Home Owner        object
Cars              float64
Commute Distance  object
Region           object
Age              float64
Purchased Bike    object
dtype: object

First 5 records:
```

Output

```

      ID Marital Status Gender      Income  Children      Education \
0  12496.0          M      F  $40,000.00        1.0      Bachelors
1  24107.0          M      M  $30,000.00        3.0  Partial College
2  14177.0          M      M  $80,000.00        5.0  Partial College
3  24381.0          S      M  $70,000.00        0.0      Bachelors
4  25597.0          S      M  $30,000.00        0.0      Bachelors

      Occupation Home Owner  Cars  Commute Distance  Region  Age \
0  Skilled Manual      Yes    0.0      0-1 Miles  Europe  42.0
1      Clerical      Yes    1.0      0-1 Miles  Europe  43.0
2    Professional      No    2.0      2-5 Miles  Europe  60.0
3    Professional      Yes    1.0      5-10 Miles  Pacific  41.0
4      Clerical      No    0.0      0-1 Miles  Europe  36.0

      Purchased Bike
0                No
1                No
2                No
3                Yes
4                Yes
```

Code

```
df.tail()
```

Output

```

      ID Marital Status Gender      Income  Children      Education \
1023  22400.0          M      M  $10,000.00        0.0  Partial College
1024  20942.0          S      F  $20,000.00        0.0      High School
1025  18484.0          S      M  $80,000.00        2.0      High School
1026      NaN          NaN    NaN          NaN        NaN          NaN
1027      0.0          NaN    NaN          NaN        NaN          NaN

      Occupation Home Owner  Cars  Commute Distance  Region  Age \
1023      Manual      No    1.0      0-1 Miles  Pacific  26.0
1024      Manual      No    1.0      5-10 Miles  Europe  31.0
1025  Skilled Manual      No    2.0      1-2 Miles  Pacific  50.0
1026      NaN          NaN    NaN          NaN        NaN        NaN
1027      NaN          NaN    NaN          NaN        NaN        NaN

      Purchased Bike
1023      Yes
1024      No
1025      Yes
1026      NaN
1027      NaN
```

Code

```
df.info()
```

Output

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1028 entries, 0 to 1027
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0    ID                    1027 non-null   float64
1    Marital Status        1026 non-null   object
2    Gender                1026 non-null   object
3    Income                1026 non-null   object
4    Children              1026 non-null   float64
5    Education             1026 non-null   object
6    Occupation            1026 non-null   object
7    Home Owner           1026 non-null   object
8    Cars                  1026 non-null   float64
9    Commute Distance      1026 non-null   object
10   Region                1026 non-null   object
11   Age                   1026 non-null   float64
12   Purchased Bike        1026 non-null   object
dtypes: float64(4), object(9)
memory usage: 104.5+ KB

```

#### Code

```
df.describe()
```

#### Output

|       | ID           | Children    | Cars        | Age         |
|-------|--------------|-------------|-------------|-------------|
| count | 1027.000000  | 1026.000000 | 1026.000000 | 1026.000000 |
| mean  | 19949.752678 | 1.892788    | 1.437622    | 44.138402   |
| std   | 5366.373922  | 1.626670    | 1.125538    | 11.349282   |
| min   | 0.000000     | 0.000000    | 0.000000    | 25.000000   |
| 25%   | 15297.000000 | 0.000000    | 1.000000    | 35.000000   |
| 50%   | 19741.000000 | 2.000000    | 1.000000    | 43.000000   |
| 75%   | 24449.500000 | 3.000000    | 2.000000    | 52.000000   |
| max   | 29447.000000 | 5.000000    | 4.000000    | 89.000000   |

#### Code

```

#finding a null values or missing values
df.isnull().sum()

```

#### Output

```

ID                1
Marital Status    2
Gender            2
Income            2
Children          2
Education         2
Occupation        2
Home Owner        2
Cars              2
Commute Distance  2
Region            2
Age              2
Purchased Bike    2
dtype: int64

```

#### Code

```

df.duplicated().sum()
np.int64(0)

```

#### Output

```
np.int64(0)
```

#### Code

```

#drop duplicate
df.drop_duplicates(inplace=True)

```

#### Code

```

df1={"num": [1, 2, 2, 3, 4, 5]}
df1=pd.DataFrame(df1)
df1

```

Output

```
      num
0      1
1      2
2      2
3      3
4      4
5      5
```

Code

```
df1.duplicated().sum()
```

Output

```
np.int64(1)
```

Code

```
df1.drop_duplicates(inplace=True)
df1
```

Output

```
      num
0      1
1      2
3      3
4      4
5      5
```

Code

```
#handling missing values
df
```

Output

```
      ID Marital Status Gender      Income  Children      Education \
0    12496.0          M      F  $40,000.00        1.0      Bachelors
1    24107.0          M      M  $30,000.00        3.0  Partial College
2    14177.0          M      M  $80,000.00        5.0  Partial College
3    24381.0          S      M   $70,000.00        0.0      Bachelors
4    25597.0          S      M  $30,000.00        0.0      Bachelors
...      ...      ...      ...      ...      ...      ...
997   11809.0          M      M   $60,000.00        2.0      Bachelors
998   19664.0          S      M  $100,000.00        3.0      Bachelors
999   12121.0          S      M   $60,000.00        3.0    High School
1026      NaN      NaN      NaN      NaN      NaN      NaN
1027      0.0      NaN      NaN      NaN      NaN      NaN

      Occupation Home Owner  Cars  Commute Distance      Region  Age \
0    Skilled Manual      Yes   0.0    0-1 Miles      Europe  42.0
1      Clerical      Yes   1.0    0-1 Miles      Europe  43.0
2    Professional      No   2.0    2-5 Miles      Europe  60.0
3    Professional      Yes   1.0    5-10 Miles    Pacific  41.0
4      Clerical      No   0.0    0-1 Miles      Europe  36.0
...      ...      ...      ...      ...      ...      ...
997   Skilled Manual      Yes   0.0    0-1 Miles  North America  38.0
998      Management      No   3.0    1-2 Miles  North America  38.0
999    Professional      Yes   2.0   10+ Miles  North America  53.0
1026      NaN      NaN      NaN      NaN      NaN      NaN
1027      NaN      NaN      NaN      NaN      NaN      NaN

      Purchased Bike
0              No
1              No
2              No
3              Yes
4              Yes
...      ...
997              Yes
998              No
999              Yes
1026            NaN
1027            NaN

[1002 rows x 13 columns]
```

Code

```
#removing null values
#df.dropna(inplace=True)
df
```

#### Output

```
      ID Marital Status Gender      Income  Children      Education \
0    12496.0          M      F  $40,000.00        1.0      Bachelors
1    24107.0          M      M  $30,000.00        3.0  Partial College
2    14177.0          M      M  $80,000.00        5.0  Partial College
3    24381.0          S      M  $70,000.00        0.0      Bachelors
4    25597.0          S      M  $30,000.00        0.0      Bachelors
...      ...      ...      ...      ...      ...
997   11809.0          M      M  $60,000.00        2.0      Bachelors
998   19664.0          S      M  $100,000.00        3.0      Bachelors
999   12121.0          S      M  $60,000.00        3.0      High School
1026      NaN          NaN      NaN          NaN      NaN          NaN
1027      0.0          NaN      NaN          NaN      NaN          NaN

      Occupation Home Owner  Cars  Commute Distance      Region  Age \
0    Skilled Manual      Yes   0.0      0-1 Miles      Europe  42.0
1      Clerical      Yes   1.0      0-1 Miles      Europe  43.0
2    Professional      No   2.0      2-5 Miles      Europe  60.0
3    Professional      Yes   1.0      5-10 Miles     Pacific  41.0
4      Clerical      No   0.0      0-1 Miles      Europe  36.0
...      ...      ...      ...      ...      ...
997   Skilled Manual      Yes   0.0      0-1 Miles  North America  38.0
998     Management      No   3.0      1-2 Miles  North America  38.0
999    Professional      Yes   2.0     10+ Miles  North America  53.0
1026      NaN          NaN      NaN          NaN      NaN      NaN
1027      NaN          NaN      NaN          NaN      NaN      NaN

      Purchased Bike
0              No
1              No
2              No
3             Yes
4             Yes
...      ...
997             Yes
998             No
999             Yes
1026           NaN
1027           NaN

[1002 rows x 13 columns]
```

#### Code

```
#fill by zero
#df.fillna(0,inplace=True)
df
```

#### Output

|      | ID      | Marital | Status | Gender  | Income       | Children | Education       | \ |
|------|---------|---------|--------|---------|--------------|----------|-----------------|---|
| 0    | 12496.0 |         |        | M F     | \$40,000.00  | 1.0      | Bachelors       |   |
| 1    | 24107.0 |         |        | M M     | \$30,000.00  | 3.0      | Partial College |   |
| 2    | 14177.0 |         |        | M M     | \$80,000.00  | 5.0      | Partial College |   |
| 3    | 24381.0 |         |        | S M     | \$70,000.00  | 0.0      | Bachelors       |   |
| 4    | 25597.0 |         |        | S M     | \$30,000.00  | 0.0      | Bachelors       |   |
| ...  | ...     |         |        | ...     | ...          | ...      | ...             |   |
| 997  | 11809.0 |         |        | M M     | \$60,000.00  | 2.0      | Bachelors       |   |
| 998  | 19664.0 |         |        | S M     | \$100,000.00 | 3.0      | Bachelors       |   |
| 999  | 12121.0 |         |        | S M     | \$60,000.00  | 3.0      | High School     |   |
| 1026 | NaN     |         |        | NaN NaN | NaN NaN      | NaN NaN  | NaN NaN         |   |
| 1027 | 0.0     |         |        | NaN NaN | NaN NaN      | NaN NaN  | NaN NaN         |   |

|      | Occupation     | Home | Owner   | Cars    | Commute | Distance   | Region        | Age     | \ |
|------|----------------|------|---------|---------|---------|------------|---------------|---------|---|
| 0    | Skilled Manual |      | Yes     | 0.0     |         | 0-1 Miles  | Europe        | 42.0    |   |
| 1    | Clerical       |      | Yes     | 1.0     |         | 0-1 Miles  | Europe        | 43.0    |   |
| 2    | Professional   |      | No      | 2.0     |         | 2-5 Miles  | Europe        | 60.0    |   |
| 3    | Professional   |      | Yes     | 1.0     |         | 5-10 Miles | Pacific       | 41.0    |   |
| 4    | Clerical       |      | No      | 0.0     |         | 0-1 Miles  | Europe        | 36.0    |   |
| ...  | ...            |      | ...     | ...     |         | ...        | ...           | ...     |   |
| 997  | Skilled Manual |      | Yes     | 0.0     |         | 0-1 Miles  | North America | 38.0    |   |
| 998  | Management     |      | No      | 3.0     |         | 1-2 Miles  | North America | 38.0    |   |
| 999  | Professional   |      | Yes     | 2.0     |         | 10+ Miles  | North America | 53.0    |   |
| 1026 | NaN            |      | NaN NaN | NaN NaN |         | NaN NaN    | NaN NaN       | NaN NaN |   |
| 1027 | NaN            |      | NaN NaN | NaN NaN |         | NaN NaN    | NaN NaN       | NaN NaN |   |

Purchased Bike

|      |     |
|------|-----|
| 0    | No  |
| 1    | No  |
| 2    | No  |
| 3    | Yes |
| 4    | Yes |
| ...  | ... |
| 997  | Yes |
| 998  | No  |
| 999  | Yes |
| 1026 | NaN |
| 1027 | NaN |

[1002 rows x 13 columns]

## Code

```
#replace by means value
df['Age'].fillna(df['Age'].mean(),inplace=True)
df['Children'].fillna(df['Children'].mean(),inplace=True)
df['ID'].fillna(df['ID'].mean(),inplace=True)
```

## Output

/tmp/ipykernel\_661/342015738.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.  
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Age'].fillna(df['Age'].mean(),inplace=True)
/tmp/ipykernel_661/342015738.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.  
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['Children'].fillna(df['Children'].mean(),inplace=True)
/tmp/ipykernel_661/342015738.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.  
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['ID'].fillna(df['ID'].mean(),inplace=True)
```

## Code

```
print(df["Age"].mean())
print(df["Children"].mean())
print(df["ID"].mean())
df
```

#### Output

```
44.163000000000004
1.8980000000000001
19946.045954045952
```

#### Output

```

      ID Marital Status Gender      Income  Children \
0    12496.000000          M      F  $40,000.00      1.000
1    24107.000000          M      M  $30,000.00      3.000
2    14177.000000          M      M  $80,000.00      5.000
3    24381.000000          S      M  $70,000.00      0.000
4    25597.000000          S      M  $30,000.00      0.000
...      ...      ...      ...      ...
997   11809.000000          M      M  $60,000.00      2.000
998   19664.000000          S      M  $100,000.00     3.000
999   12121.000000          S      M  $60,000.00      3.000
1026  19946.045954         NaN     NaN         NaN      1.898
1027    0.000000         NaN     NaN         NaN      1.898

      Education      Occupation Home Owner  Cars  Commute Distance \
0      Bachelors  Skilled Manual      Yes    0.0      0-1 Miles
1  Partial College      Clerical      Yes    1.0      0-1 Miles
2  Partial College  Professional      No     2.0      2-5 Miles
3      Bachelors  Professional      Yes    1.0      5-10 Miles
4      Bachelors      Clerical      No     0.0      0-1 Miles
...      ...      ...      ...      ...      ...
997      Bachelors  Skilled Manual      Yes    0.0      0-1 Miles
998      Bachelors      Management      No    3.0      1-2 Miles
999    High School  Professional      Yes    2.0      10+ Miles
1026         NaN         NaN         NaN    NaN         NaN
1027         NaN         NaN         NaN    NaN         NaN

      Region      Age Purchased Bike
0      Europe  42.000          No
1      Europe  43.000          No
2      Europe  60.000          No
3    Pacific  41.000          Yes
4      Europe  36.000          Yes
...      ...      ...      ...
997 North America  38.000          Yes
998 North America  38.000          No
999 North America  53.000          Yes
1026         NaN  44.163         NaN
1027         NaN  44.163         NaN

[1002 rows x 13 columns]
```

#### Code

```
#data analysis-groupby
df.groupby("Cars")["Age"].mean()
```

#### Output

```
Cars
0.0    41.267206
1.0    41.438202
2.0    47.792398
3.0    46.035294
4.0    44.881356
Name: Age, dtype: float64
```

#### Code

```
pr=pd.read_csv("/content/Bike Sales Data.csv")
pr
```

#### Output

|      | ID      | Marital Status | Gender | Income      | Children | Education       | \   |
|------|---------|----------------|--------|-------------|----------|-----------------|-----|
| 0    | 12496.0 |                | M F    | \$40,000.00 | 1.0      | Bachelors       |     |
| 1    | 24107.0 |                | M M    | \$30,000.00 | 3.0      | Partial College |     |
| 2    | 14177.0 |                | M M    | \$80,000.00 | 5.0      | Partial College |     |
| 3    | 24381.0 |                | S M    | \$70,000.00 | 0.0      | Bachelors       |     |
| 4    | 25597.0 |                | S M    | \$30,000.00 | 0.0      | Bachelors       |     |
| ...  | ...     | ...            | ...    | ...         | ...      | ...             | ... |
| 1023 | 22400.0 |                | M M    | \$10,000.00 | 0.0      | Partial College |     |
| 1024 | 20942.0 |                | S F    | \$20,000.00 | 0.0      | High School     |     |
| 1025 | 18484.0 |                | S M    | \$80,000.00 | 2.0      | High School     |     |
| 1026 | NaN     | NaN            | NaN    | NaN         | NaN      | NaN             | NaN |
| 1027 | 0.0     | NaN            | NaN    | NaN         | NaN      | NaN             | NaN |

|      | Occupation     | Home Owner | Cars | Commute Distance | Region  | Age  | \   |
|------|----------------|------------|------|------------------|---------|------|-----|
| 0    | Skilled Manual | Yes        | 0.0  | 0-1 Miles        | Europe  | 42.0 |     |
| 1    | Clerical       | Yes        | 1.0  | 0-1 Miles        | Europe  | 43.0 |     |
| 2    | Professional   | No         | 2.0  | 2-5 Miles        | Europe  | 60.0 |     |
| 3    | Professional   | Yes        | 1.0  | 5-10 Miles       | Pacific | 41.0 |     |
| 4    | Clerical       | No         | 0.0  | 0-1 Miles        | Europe  | 36.0 |     |
| ...  | ...            | ...        | ...  | ...              | ...     | ...  | ... |
| 1023 | Manual         | No         | 1.0  | 0-1 Miles        | Pacific | 26.0 |     |
| 1024 | Manual         | No         | 1.0  | 5-10 Miles       | Europe  | 31.0 |     |
| 1025 | Skilled Manual | No         | 2.0  | 1-2 Miles        | Pacific | 50.0 |     |
| 1026 | NaN            | NaN        | NaN  | NaN              | NaN     | NaN  | NaN |
| 1027 | NaN            | NaN        | NaN  | NaN              | NaN     | NaN  | NaN |

Purchased Bike

|      |     |
|------|-----|
| 0    | No  |
| 1    | No  |
| 2    | No  |
| 3    | Yes |
| 4    | Yes |
| ...  | ... |
| 1023 | Yes |
| 1024 | No  |
| 1025 | Yes |
| 1026 | NaN |
| 1027 | NaN |

[1028 rows x 13 columns]

## Code

```
pr.groupby("ID")["Income"].count()
```

## Output

```
ID
0.0      0
11000.0   1
11047.0   1
11061.0   1
11090.0   1
..
29337.0   1
29355.0   1
29380.0   1
29424.0   1
29447.0   1
Name: Income, Length: 1001, dtype: int64
```

## Code

```
#features engineering
#create a features in a existing data
def grade(Age):
    if Age>=80:
        return"A"
    elif Age>=70:
        return"B"
    else:
        return"C"
df["Grade"]=df["Age"].apply(grade)
df
```

## Output

|      | ID           | Marital | Status | Gender | Income | Children     | \     |
|------|--------------|---------|--------|--------|--------|--------------|-------|
| 0    | 12496.000000 |         |        | M      | F      | \$40,000.00  | 1.000 |
| 1    | 24107.000000 |         |        | M      | M      | \$30,000.00  | 3.000 |
| 2    | 14177.000000 |         |        | M      | M      | \$80,000.00  | 5.000 |
| 3    | 24381.000000 |         |        | S      | M      | \$70,000.00  | 0.000 |
| 4    | 25597.000000 |         |        | S      | M      | \$30,000.00  | 0.000 |
| ...  | ...          |         |        | ...    | ...    | ...          | ...   |
| 997  | 11809.000000 |         |        | M      | M      | \$60,000.00  | 2.000 |
| 998  | 19664.000000 |         |        | S      | M      | \$100,000.00 | 3.000 |
| 999  | 12121.000000 |         |        | S      | M      | \$60,000.00  | 3.000 |
| 1026 | 19946.045954 |         |        | NaN    | NaN    | NaN          | 1.898 |
| 1027 | 0.000000     |         |        | NaN    | NaN    | NaN          | 1.898 |

|      | Education       | Occupation     | Home | Owner | Cars | Commute | Distance   | \ |
|------|-----------------|----------------|------|-------|------|---------|------------|---|
| 0    | Bachelors       | Skilled Manual |      | Yes   | 0.0  |         | 0-1 Miles  |   |
| 1    | Partial College | Clerical       |      | Yes   | 1.0  |         | 0-1 Miles  |   |
| 2    | Partial College | Professional   |      | No    | 2.0  |         | 2-5 Miles  |   |
| 3    | Bachelors       | Professional   |      | Yes   | 1.0  |         | 5-10 Miles |   |
| 4    | Bachelors       | Clerical       |      | No    | 0.0  |         | 0-1 Miles  |   |
| ...  | ...             | ...            |      | ...   | ...  |         | ...        |   |
| 997  | Bachelors       | Skilled Manual |      | Yes   | 0.0  |         | 0-1 Miles  |   |
| 998  | Bachelors       | Management     |      | No    | 3.0  |         | 1-2 Miles  |   |
| 999  | High School     | Professional   |      | Yes   | 2.0  |         | 10+ Miles  |   |
| 1026 | NaN             | NaN            |      | NaN   | NaN  |         | NaN        |   |
| 1027 | NaN             | NaN            |      | NaN   | NaN  |         | NaN        |   |

|      | Region        | Age    | Purchased | Bike | Grade |
|------|---------------|--------|-----------|------|-------|
| 0    | Europe        | 42.000 |           | No   | C     |
| 1    | Europe        | 43.000 |           | No   | C     |
| 2    | Europe        | 60.000 |           | No   | C     |
| 3    | Pacific       | 41.000 |           | Yes  | C     |
| 4    | Europe        | 36.000 |           | Yes  | C     |
| ...  | ...           | ...    |           | ...  | ...   |
| 997  | North America | 38.000 |           | Yes  | C     |
| 998  | North America | 38.000 |           | No   | C     |
| 999  | North America | 53.000 |           | Yes  | C     |
| 1026 | NaN           | 44.163 |           | NaN  | C     |
| 1027 | NaN           | 44.163 |           | NaN  | C     |

[1002 rows x 14 columns]

## Code

```
#product data
data=pd.read_csv("/content/Bike Sales Data.csv")
data
```

## Output

|      | ID      | Marital | Status | Gender | Income | Children    | Education | \               |
|------|---------|---------|--------|--------|--------|-------------|-----------|-----------------|
| 0    | 12496.0 |         |        | M      | F      | \$40,000.00 | 1.0       | Bachelors       |
| 1    | 24107.0 |         |        | M      | M      | \$30,000.00 | 3.0       | Partial College |
| 2    | 14177.0 |         |        | M      | M      | \$80,000.00 | 5.0       | Partial College |
| 3    | 24381.0 |         |        | S      | M      | \$70,000.00 | 0.0       | Bachelors       |
| 4    | 25597.0 |         |        | S      | M      | \$30,000.00 | 0.0       | Bachelors       |
| ...  | ...     |         |        | ...    | ...    | ...         | ...       | ...             |
| 1023 | 22400.0 |         |        | M      | M      | \$10,000.00 | 0.0       | Partial College |
| 1024 | 20942.0 |         |        | S      | F      | \$20,000.00 | 0.0       | High School     |
| 1025 | 18484.0 |         |        | S      | M      | \$80,000.00 | 2.0       | High School     |
| 1026 | NaN     |         |        | NaN    | NaN    | NaN         | NaN       | NaN             |
| 1027 | 0.0     |         |        | NaN    | NaN    | NaN         | NaN       | NaN             |

|      | Occupation     | Home | Owner | Cars | Commute | Distance   | Region  | Age  | \ |
|------|----------------|------|-------|------|---------|------------|---------|------|---|
| 0    | Skilled Manual |      | Yes   | 0.0  |         | 0-1 Miles  | Europe  | 42.0 |   |
| 1    | Clerical       |      | Yes   | 1.0  |         | 0-1 Miles  | Europe  | 43.0 |   |
| 2    | Professional   |      | No    | 2.0  |         | 2-5 Miles  | Europe  | 60.0 |   |
| 3    | Professional   |      | Yes   | 1.0  |         | 5-10 Miles | Pacific | 41.0 |   |
| 4    | Clerical       |      | No    | 0.0  |         | 0-1 Miles  | Europe  | 36.0 |   |
| ...  | ...            |      | ...   | ...  |         | ...        | ...     | ...  |   |
| 1023 | Manual         |      | No    | 1.0  |         | 0-1 Miles  | Pacific | 26.0 |   |
| 1024 | Manual         |      | No    | 1.0  |         | 5-10 Miles | Europe  | 31.0 |   |
| 1025 | Skilled Manual |      | No    | 2.0  |         | 1-2 Miles  | Pacific | 50.0 |   |
| 1026 | NaN            |      | NaN   | NaN  |         | NaN        | NaN     | NaN  |   |
| 1027 | NaN            |      | NaN   | NaN  |         | NaN        | NaN     | NaN  |   |

|      | Purchased | Bike |
|------|-----------|------|
| 0    | No        |      |
| 1    | No        |      |
| 2    | No        |      |
| 3    | Yes       |      |
| 4    | Yes       |      |
| ...  | ...       |      |
| 1023 | Yes       |      |
| 1024 | No        |      |
| 1025 | Yes       |      |
| 1026 | NaN       |      |
| 1027 | NaN       |      |

[1028 rows x 13 columns]

|                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Code                                                                                                                                                                  |
| <pre>#groupby data.groupby("Gender")["Age"].mean()</pre>                                                                                                              |
| Output                                                                                                                                                                |
| <pre>Gender F    44.123752 M    44.152381 Name: Age, dtype: float64</pre>                                                                                             |
| Code                                                                                                                                                                  |
| <pre>#groupby data.groupby("Gender")["Income"].sum()</pre>                                                                                                            |
| Output                                                                                                                                                                |
| <pre>Gender F    \$40,000.00\$10,000.00\$30,000.00\$90,000.00\$10,00... M    \$30,000.00\$80,000.00\$70,000.00\$30,000.00\$160,0... Name: Income, dtype: object</pre> |
| Code                                                                                                                                                                  |
| <pre>#standard deviation data.groupby("Gender")["Age"].std()</pre>                                                                                                    |
| Output                                                                                                                                                                |
| <pre>Gender F    11.365239 M    11.344855 Name: Age, dtype: float64</pre>                                                                                             |
| Code                                                                                                                                                                  |
| <pre>#merge Rating={"Rating":[4,5,4,3,3,4,5,2,3,4,4]} Rating=pd.DataFrame(Rating) Rating</pre>                                                                        |
| Output                                                                                                                                                                |
| <pre>      Rating 0         4 1         5 2         4 3         3 4         3 5         4 6         5 7         2 8         3 9         4 10        4</pre>           |
| Code                                                                                                                                                                  |
| <pre>#merge data.merge(Rating, left_index=True, right_index=True)</pre>                                                                                               |
| Output                                                                                                                                                                |

|    | ID      | Marital Status | Gender | Income       | Children | Education \         |
|----|---------|----------------|--------|--------------|----------|---------------------|
| 0  | 12496.0 |                | M F    | \$40,000.00  | 1.0      | Bachelors           |
| 1  | 24107.0 |                | M M    | \$30,000.00  | 3.0      | Partial College     |
| 2  | 14177.0 |                | M M    | \$80,000.00  | 5.0      | Partial College     |
| 3  | 24381.0 |                | S M    | \$70,000.00  | 0.0      | Bachelors           |
| 4  | 25597.0 |                | S M    | \$30,000.00  | 0.0      | Bachelors           |
| 5  | 13507.0 |                | M F    | \$10,000.00  | 2.0      | Partial College     |
| 6  | 27974.0 |                | S M    | \$160,000.00 | 2.0      | High School         |
| 7  | 19364.0 |                | M M    | \$40,000.00  | 1.0      | Bachelors           |
| 8  | 22155.0 |                | M M    | \$20,000.00  | 2.0      | Partial High School |
| 9  | 19280.0 |                | M M    | \$120,000.00 | 2.0      | Partial College     |
| 10 | 22173.0 |                | M F    | \$30,000.00  | 3.0      | High School         |

|    | Occupation     | Home Owner | Cars | Commute Distance | Region  | Age \ |
|----|----------------|------------|------|------------------|---------|-------|
| 0  | Skilled Manual | Yes        | 0.0  | 0-1 Miles        | Europe  | 42.0  |
| 1  | Clerical       | Yes        | 1.0  | 0-1 Miles        | Europe  | 43.0  |
| 2  | Professional   | No         | 2.0  | 2-5 Miles        | Europe  | 60.0  |
| 3  | Professional   | Yes        | 1.0  | 5-10 Miles       | Pacific | 41.0  |
| 4  | Clerical       | No         | 0.0  | 0-1 Miles        | Europe  | 36.0  |
| 5  | Manual         | Yes        | 0.0  | 1-2 Miles        | Europe  | 50.0  |
| 6  | Management     | Yes        | 4.0  | 0-1 Miles        | Pacific | 33.0  |
| 7  | Skilled Manual | Yes        | 0.0  | 0-1 Miles        | Europe  | 43.0  |
| 8  | Clerical       | Yes        | 2.0  | 5-10 Miles       | Pacific | 58.0  |
| 9  | Manual         | Yes        | 1.0  | 0-1 Miles        | Europe  | 40.0  |
| 10 | Skilled Manual | No         | 2.0  | 1-2 Miles        | Pacific | 54.0  |

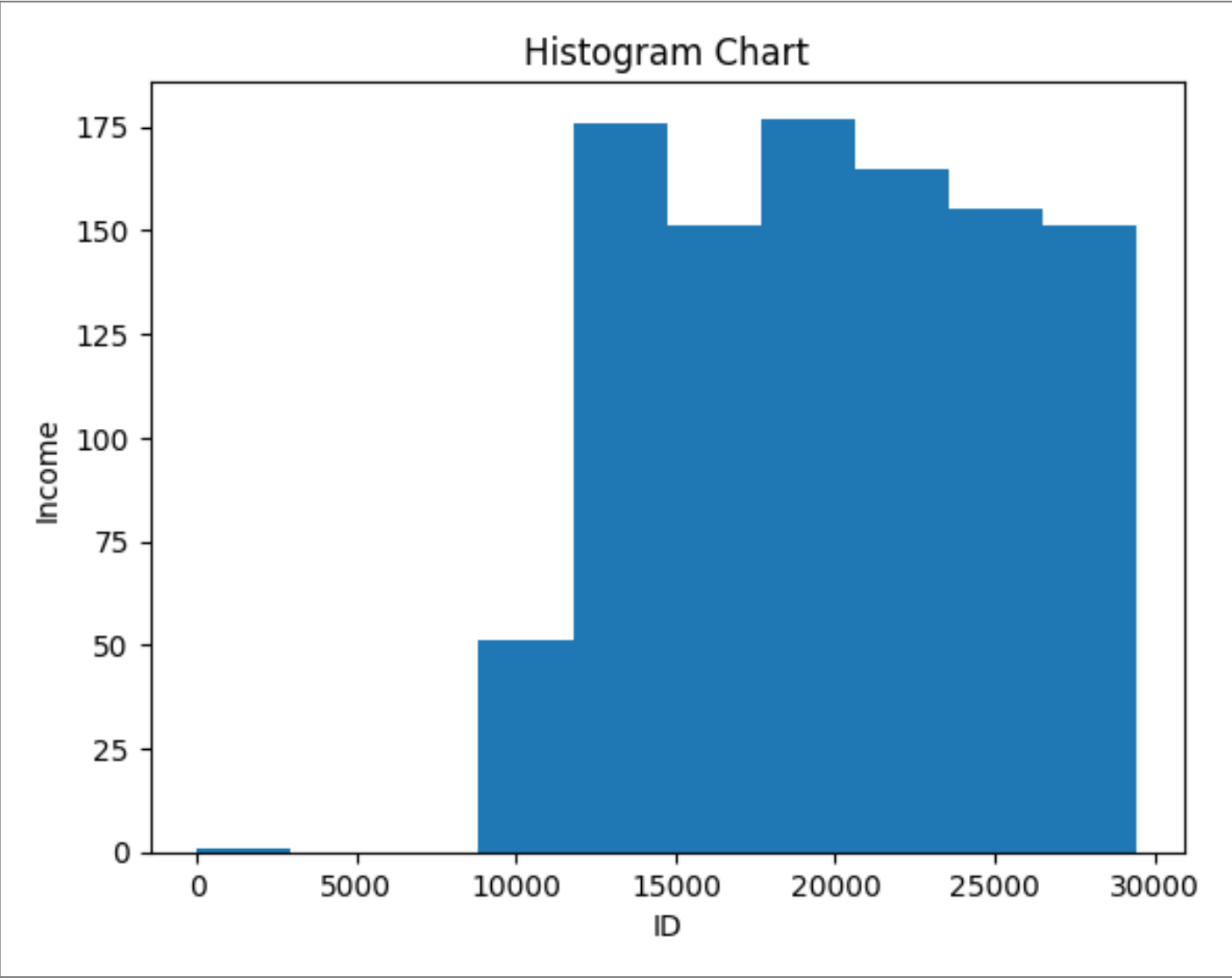
|    | Purchased Bike | Rating |
|----|----------------|--------|
| 0  | No             | 4      |
| 1  | No             | 5      |
| 2  | No             | 4      |
| 3  | Yes            | 3      |
| 4  | Yes            | 3      |
| 5  | No             | 4      |
| 6  | Yes            | 5      |
| 7  | Yes            | 2      |
| 8  | No             | 3      |
| 9  | Yes            | 4      |
| 10 | Yes            | 4      |

## Code

```
#EDA process
#univariate Analysis
#histogram chart
plt.hist(data['ID'])
plt.xlabel("ID")
plt.ylabel("Income")
plt.title("Histogram Chart")
plt.show()
```

## Output

<Figure size 640x480 with 1 Axes>



Code

```
data['ID'].mean()
```

Output

```
np.float64(19949.752677702043)
```

Code

```
data['ID'].std()
```

Output

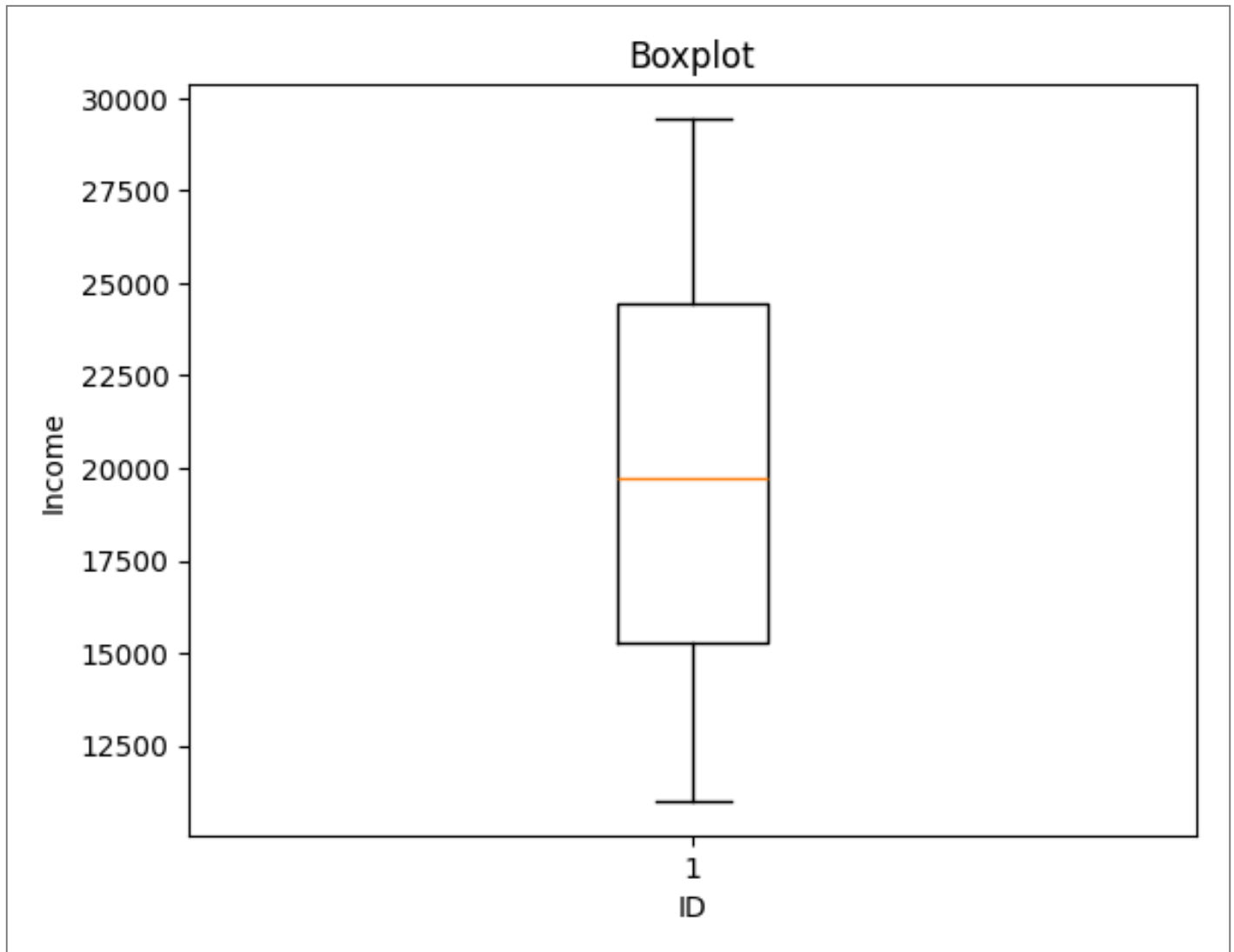
```
5366.373921693033
```

Code

```
#Boxplot
plt.boxplot(data['ID'])
plt.xlabel("ID")
plt.ylabel("Income")
plt.title("Boxplot")
plt.show()
```

Output

```
<Figure size 640x480 with 1 Axes>
```



#### Code

```
Q1 = data['ID'].quantile(0.25)
Q3 = data['ID'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = data[(data['ID'] < lower_bound) | (data['ID'] > upper_bound)]
print(f"Q1 (25th percentile): {Q1}")
print(f"Q3 (75th percentile): {Q3}")
print(f"IQR (Interquartile Range): {IQR}")
print(f"Lower Bound for Outliers: {lower_bound}")
print(f"Upper Bound for Outliers: {upper_bound}")
if not outliers.empty:
    print("\nPotential Outliers in ID:")
    display(outliers)
else:
    print("\nNo potential outliers found in ID based on the IQR method:")
```

#### Output

```
Q1 (25th percentile): 15304.75
Q3 (75th percentile): 24457.75
IQR (Interquartile Range): 9153.0
Lower Bound for Outliers: 1575.25
Upper Bound for Outliers: 38187.25
```

No potential outliers found in ID based on the IQR method:

#### Code

```
#Remove the outliers
data = data[(data['ID'] >= lower_bound) & (data['ID'] <= upper_bound)]
data
```

Output

|      | ID      | Marital | Status | Gender | Income      | Children | \ |
|------|---------|---------|--------|--------|-------------|----------|---|
| 0    | 12496.0 |         | M      | F      | \$40,000.00 | 1.0      |   |
| 1    | 24107.0 |         | M      | M      | \$30,000.00 | 3.0      |   |
| 2    | 14177.0 |         | M      | M      | \$80,000.00 | 5.0      |   |
| 3    | 24381.0 |         | S      | M      | \$70,000.00 | 0.0      |   |
| 4    | 25597.0 |         | S      | M      | \$30,000.00 | 0.0      |   |
| ...  | ...     |         | ...    | ...    | ...         | ...      |   |
| 1021 | 16466.0 |         | S      | F      | \$20,000.00 | 0.0      |   |
| 1022 | 19273.0 |         | M      | F      | \$20,000.00 | 2.0      |   |
| 1023 | 22400.0 |         | M      | M      | \$10,000.00 | 0.0      |   |
| 1024 | 20942.0 |         | S      | F      | \$20,000.00 | 0.0      |   |
| 1025 | 18484.0 |         | S      | M      | \$80,000.00 | 2.0      |   |

|      | Education           | Occupation     | Home | Owner | Cars | Commute | Distance   | \ |
|------|---------------------|----------------|------|-------|------|---------|------------|---|
| 0    | Bachelors           | Skilled Manual |      | Yes   | 0.0  |         | 0-1 Miles  |   |
| 1    | Partial College     | Clerical       |      | Yes   | 1.0  |         | 0-1 Miles  |   |
| 2    | Partial College     | Professional   |      | No    | 2.0  |         | 2-5 Miles  |   |
| 3    | Bachelors           | Professional   |      | Yes   | 1.0  |         | 5-10 Miles |   |
| 4    | Bachelors           | Clerical       |      | No    | 0.0  |         | 0-1 Miles  |   |
| ...  | ...                 | ...            |      | ...   | ...  |         | ...        |   |
| 1021 | Partial High School | Manual         |      | No    | 2.0  |         | 0-1 Miles  |   |
| 1022 | Partial College     | Manual         |      | Yes   | 0.0  |         | 0-1 Miles  |   |
| 1023 | Partial College     | Manual         |      | No    | 1.0  |         | 0-1 Miles  |   |
| 1024 | High School         | Manual         |      | No    | 1.0  |         | 5-10 Miles |   |
| 1025 | High School         | Skilled Manual |      | No    | 2.0  |         | 1-2 Miles  |   |

|      | Region  | Age  | Purchased | Bike |
|------|---------|------|-----------|------|
| 0    | Europe  | 42.0 |           | No   |
| 1    | Europe  | 43.0 |           | No   |
| 2    | Europe  | 60.0 |           | No   |
| 3    | Pacific | 41.0 |           | Yes  |
| 4    | Europe  | 36.0 |           | Yes  |
| ...  | ...     | ...  |           | ...  |
| 1021 | Europe  | 32.0 |           | Yes  |
| 1022 | Europe  | 63.0 |           | No   |
| 1023 | Pacific | 26.0 |           | Yes  |
| 1024 | Europe  | 31.0 |           | No   |
| 1025 | Pacific | 50.0 |           | Yes  |

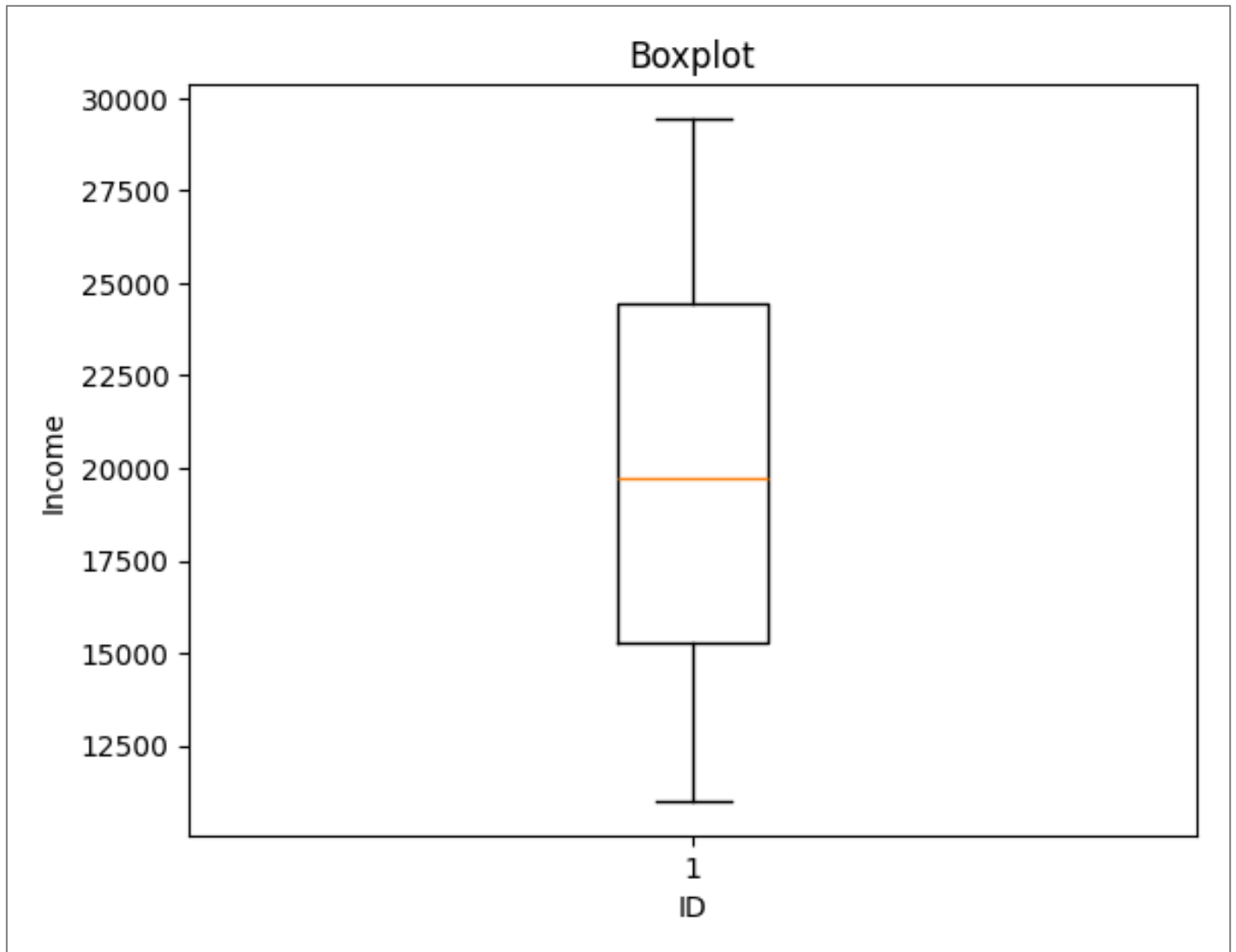
[1026 rows x 13 columns]

Code

```
#boxplot
plt.boxplot(data['ID'])
plt.xlabel("ID")
plt.ylabel("Income")
plt.title("Boxplot")
plt.show()
```

Output

<Figure size 640x480 with 1 Axes>



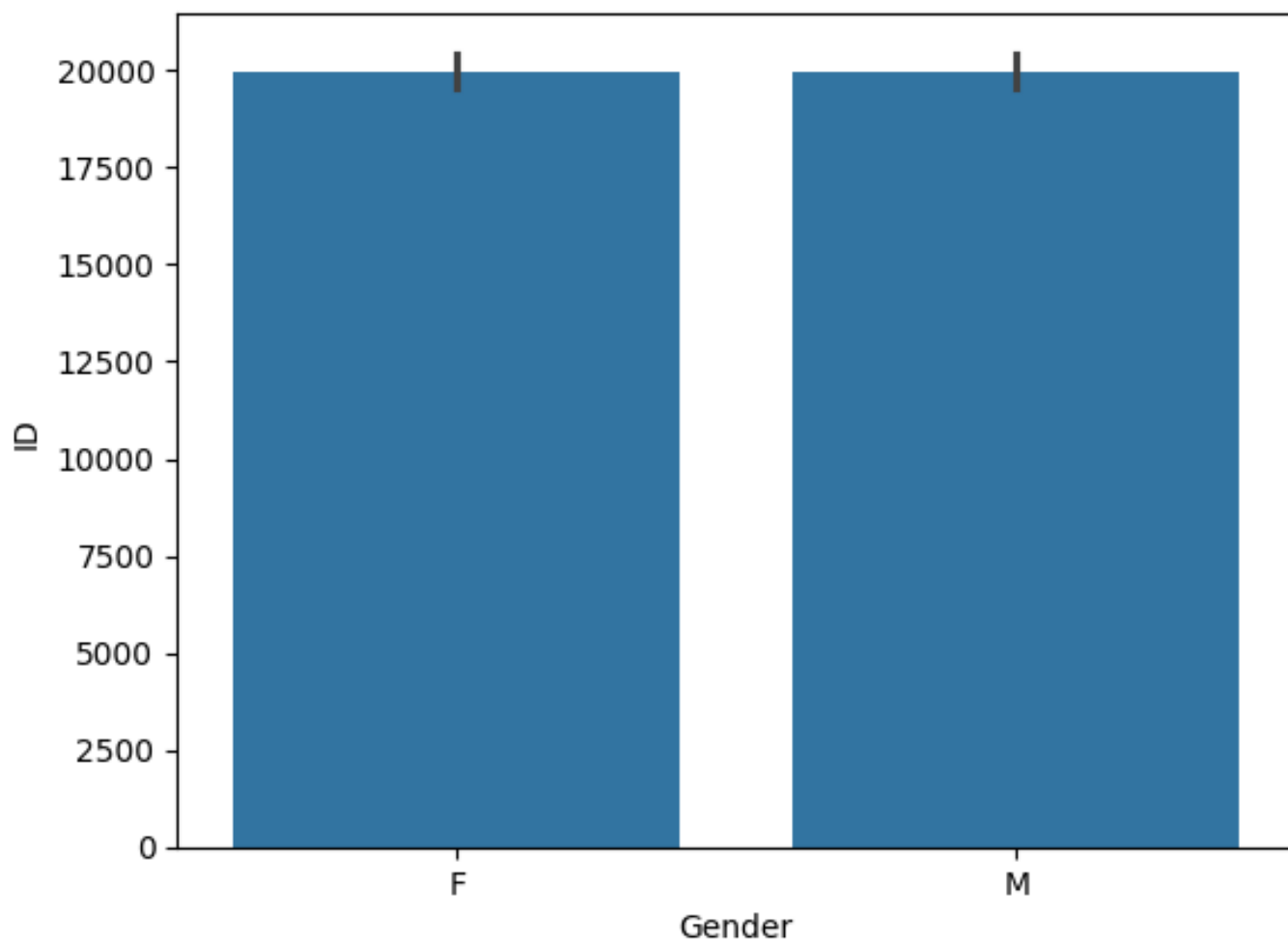
#### Code

```
#Bivariate Analysis
#Bar chart
sns.barplot(x=data['Gender'],y=data['ID'])
plt.xlabel("Gender")
plt.ylabel("ID")
plt.title("Bar Chart")
plt.show()
```

#### Output

<Figure size 640x480 with 1 Axes>

Bar Chart



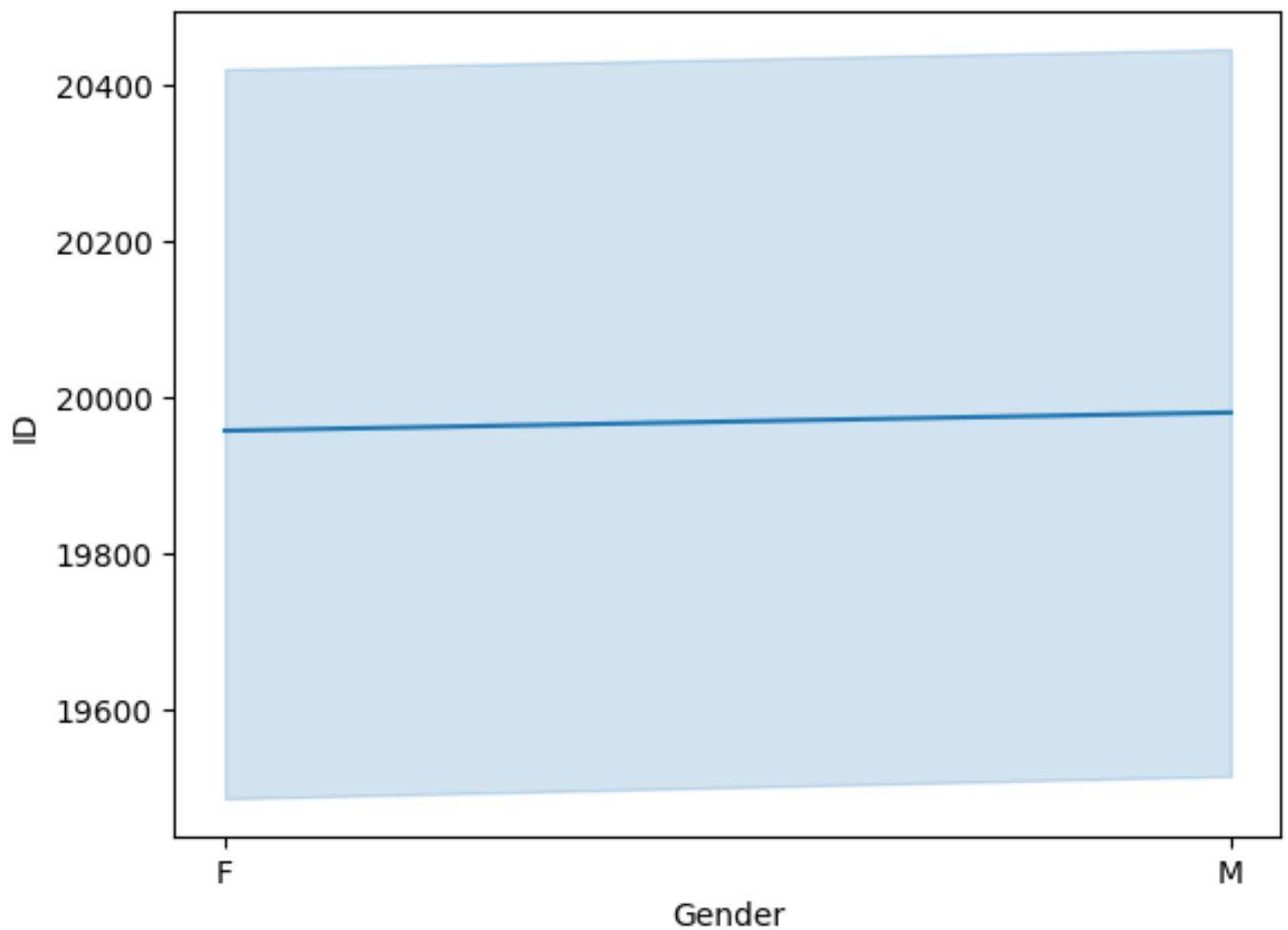
Code

```
#line plot
sns.lineplot(x=data['Gender'],y=data['ID'])
plt.xlabel("Gender")
plt.ylabel("ID")
plt.title("Line Plot")
plt.show()
```

Output

<Figure size 640x480 with 1 Axes>

Line Plot



Code

```
d1 = data.drop(columns=["ID","Income","Region","Gender","Marital Status"])
d1
```

Output

```

   Children  Education  Occupation  Home Owner  Cars  \
0         1.0    Bachelors  Skilled Manual      Yes    0.0
1         3.0  Partial College    Clerical      Yes    1.0
2         5.0  Partial College  Professional    No     2.0
3         0.0    Bachelors  Professional      Yes    1.0
4         0.0    Bachelors    Clerical      No     0.0
...      ...      ...      ...      ...      ...
1021        0.0  Partial High School    Manual      No    2.0
1022        2.0  Partial College    Manual      Yes    0.0
1023        0.0  Partial College    Manual      No    1.0
1024        0.0    High School    Manual      No    1.0
1025        2.0    High School  Skilled Manual      No    2.0
```

```

   Commute Distance  Age  Purchased  Bike
0         0-1 Miles  42.0         No
1         0-1 Miles  43.0         No
2         2-5 Miles  60.0         No
3         5-10 Miles  41.0         Yes
4         0-1 Miles  36.0         Yes
...      ...      ...      ...
1021        0-1 Miles  32.0         Yes
1022        0-1 Miles  63.0         No
1023        0-1 Miles  26.0         Yes
1024        5-10 Miles  31.0         No
1025        1-2 Miles  50.0         Yes
```

[1026 rows x 8 columns]

Code

```
df.select_dtypes(include=[np.number]).corr()
```

### Output

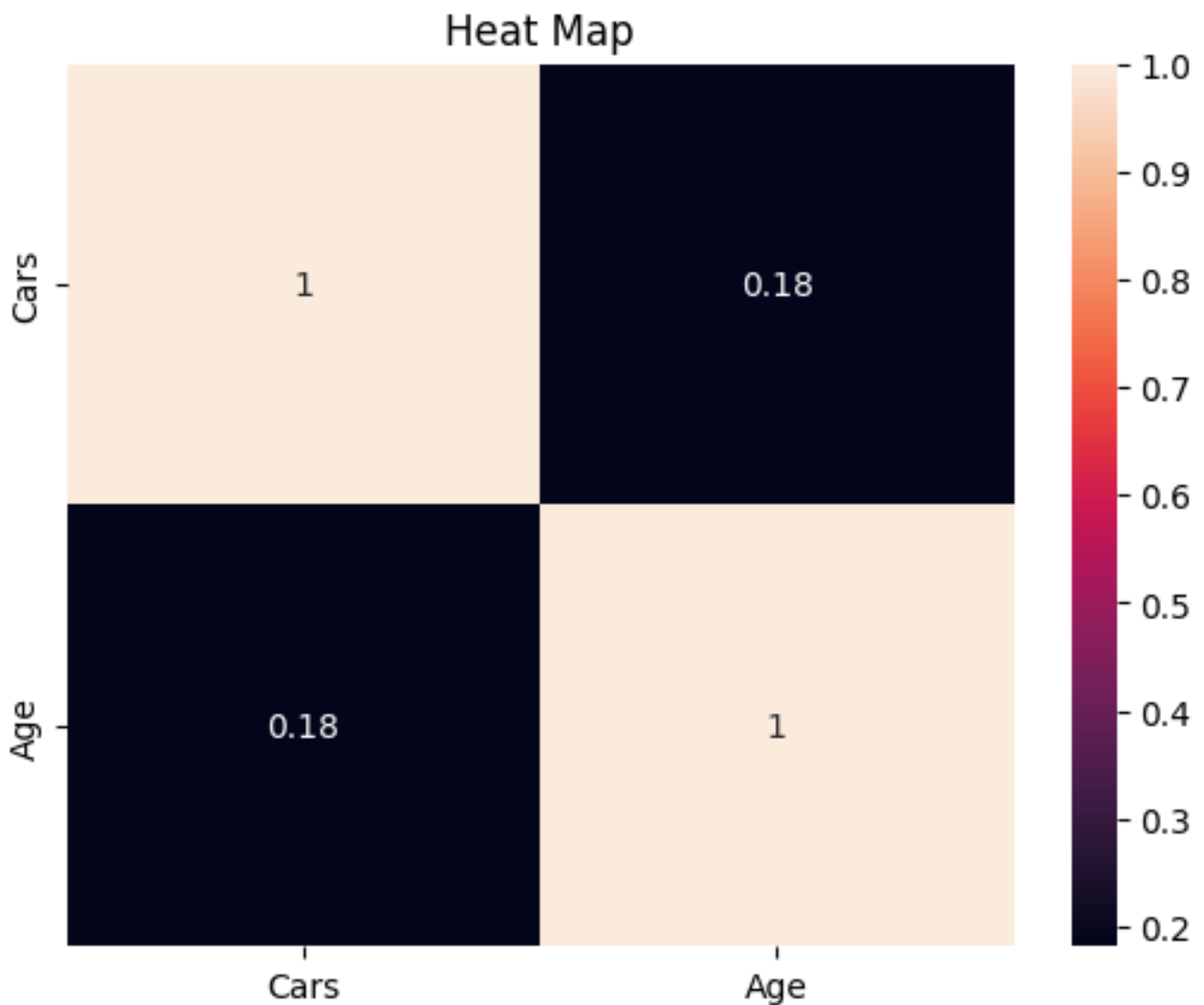
|          | ID        | Children  | Cars     | Age       |
|----------|-----------|-----------|----------|-----------|
| ID       | 1.000000  | -0.030773 | 0.032816 | -0.049564 |
| Children | -0.030773 | 1.000000  | 0.263360 | 0.529743  |
| Cars     | 0.032816  | 0.263360  | 1.000000 | 0.188040  |
| Age      | -0.049564 | 0.529743  | 0.188040 | 1.000000  |

### Code

```
#heat map
sns.heatmap(d1[['Cars', 'Age']].corr(), annot=True)
plt.title("Heat Map")
plt.show()
```

### Output

<Figure size 640x480 with 2 Axes>



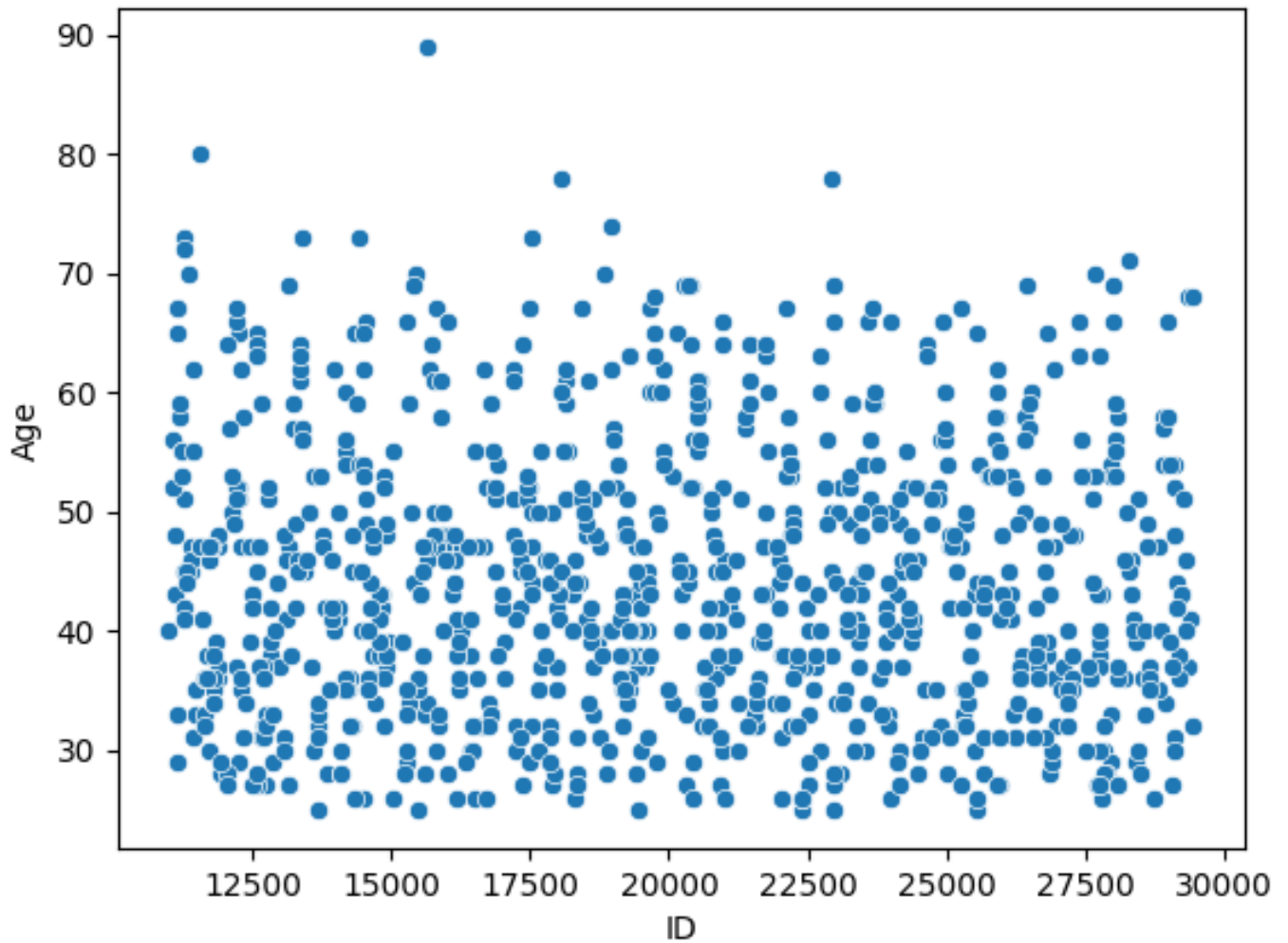
### Code

```
#Scatter plot
sns.scatterplot(x=data['ID'], y=data['Age'])
plt.xlabel("ID")
plt.ylabel("Age")
plt.title("Scatter Plot")
plt.show()
```

### Output

<Figure size 640x480 with 1 Axes>

Scatter Plot

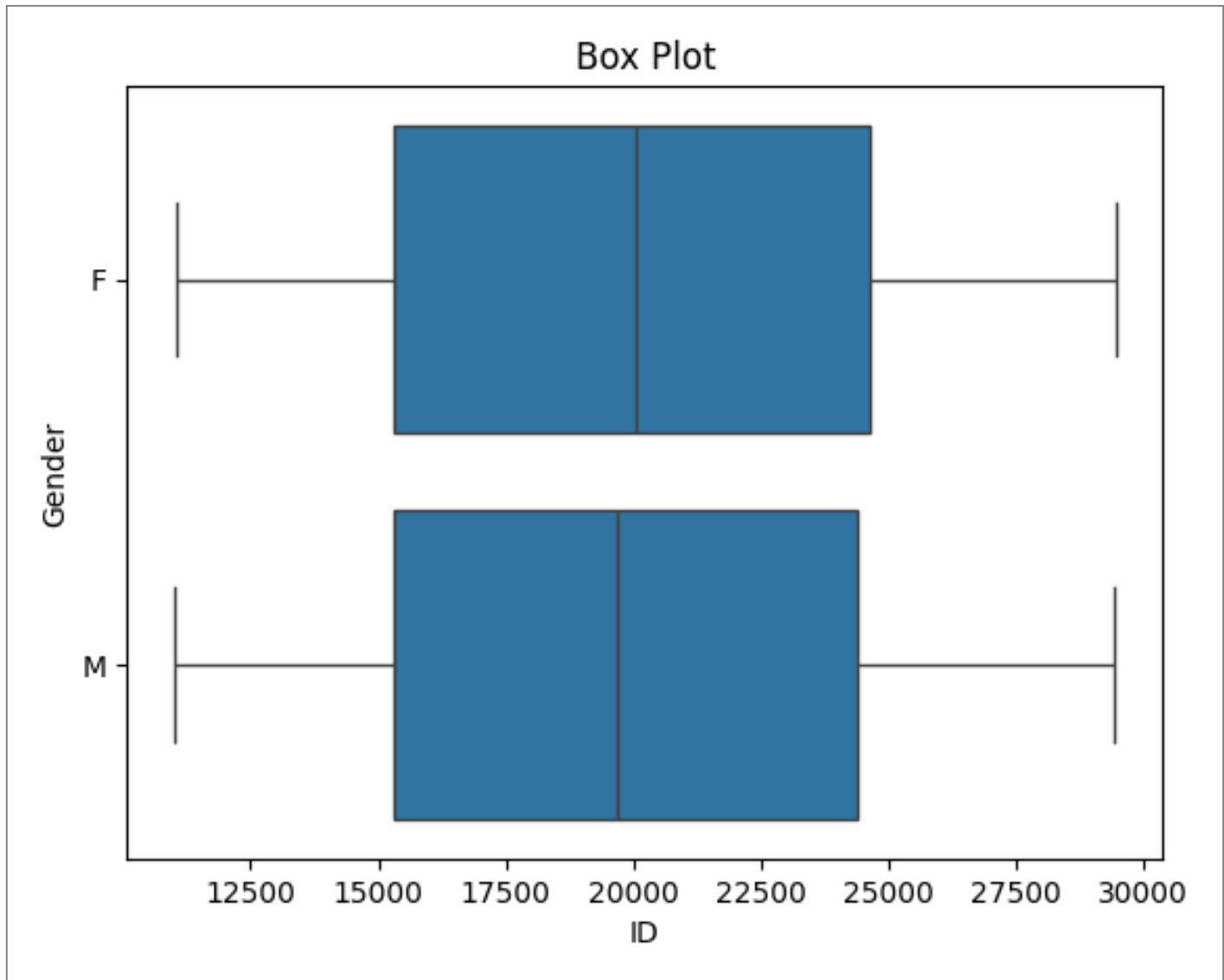


#### Code

```
#bivariate boxplot
sns.boxplot(x=data['ID'],y=data['Gender'])
plt.xlabel("ID")
plt.ylabel("Gender")
plt.title("Box Plot")
plt.show()
```

#### Output

<Figure size 640x480 with 1 Axes>



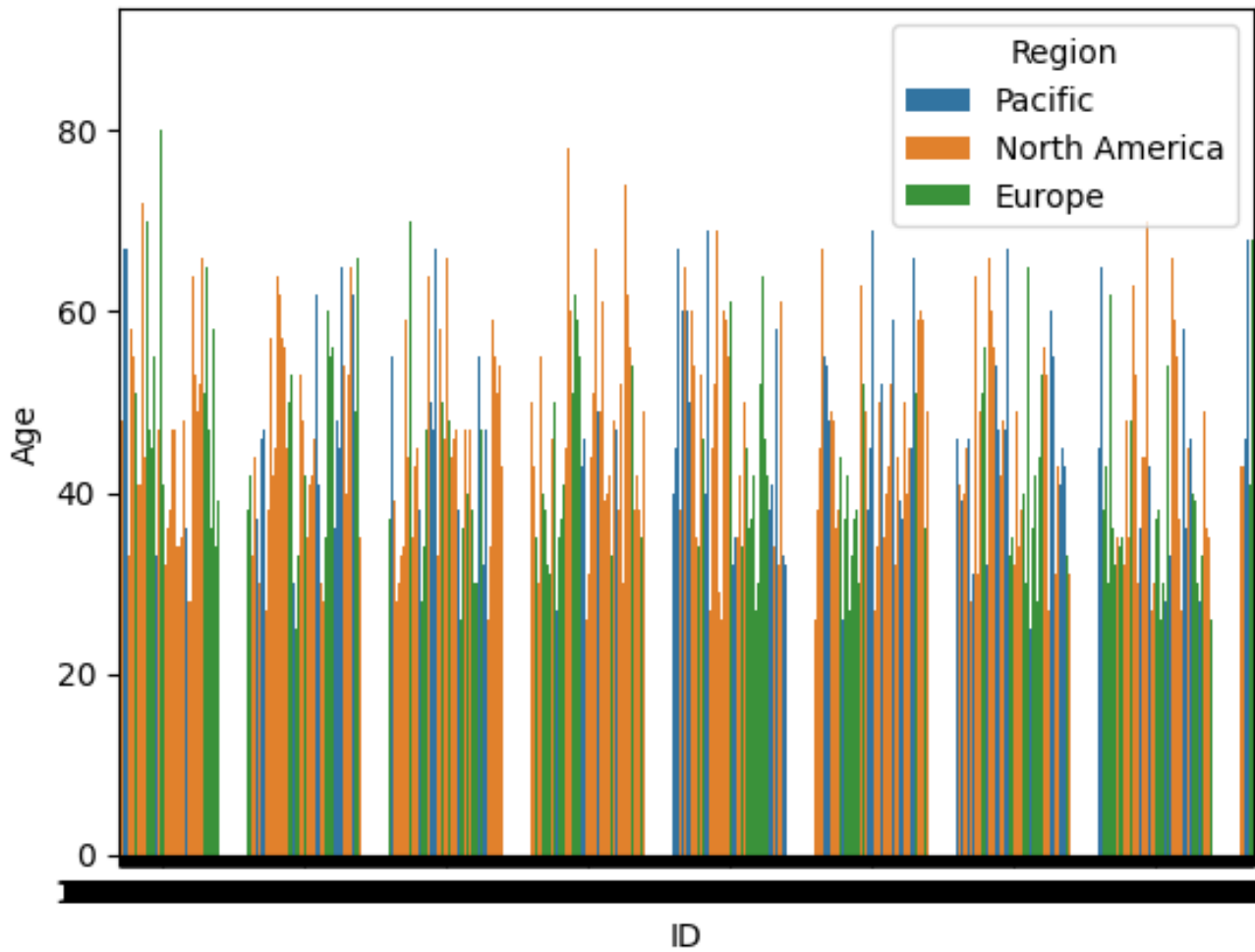
#### Code

```
#multivariate analysis
sns.barplot(x=data['ID'],y=data['Age'],hue=data['Region'])
plt.xlabel("ID")
plt.ylabel("Age")
plt.title("Bar Chart")
plt.show()
```

#### Output

<Figure size 640x480 with 1 Axes>

Bar Chart



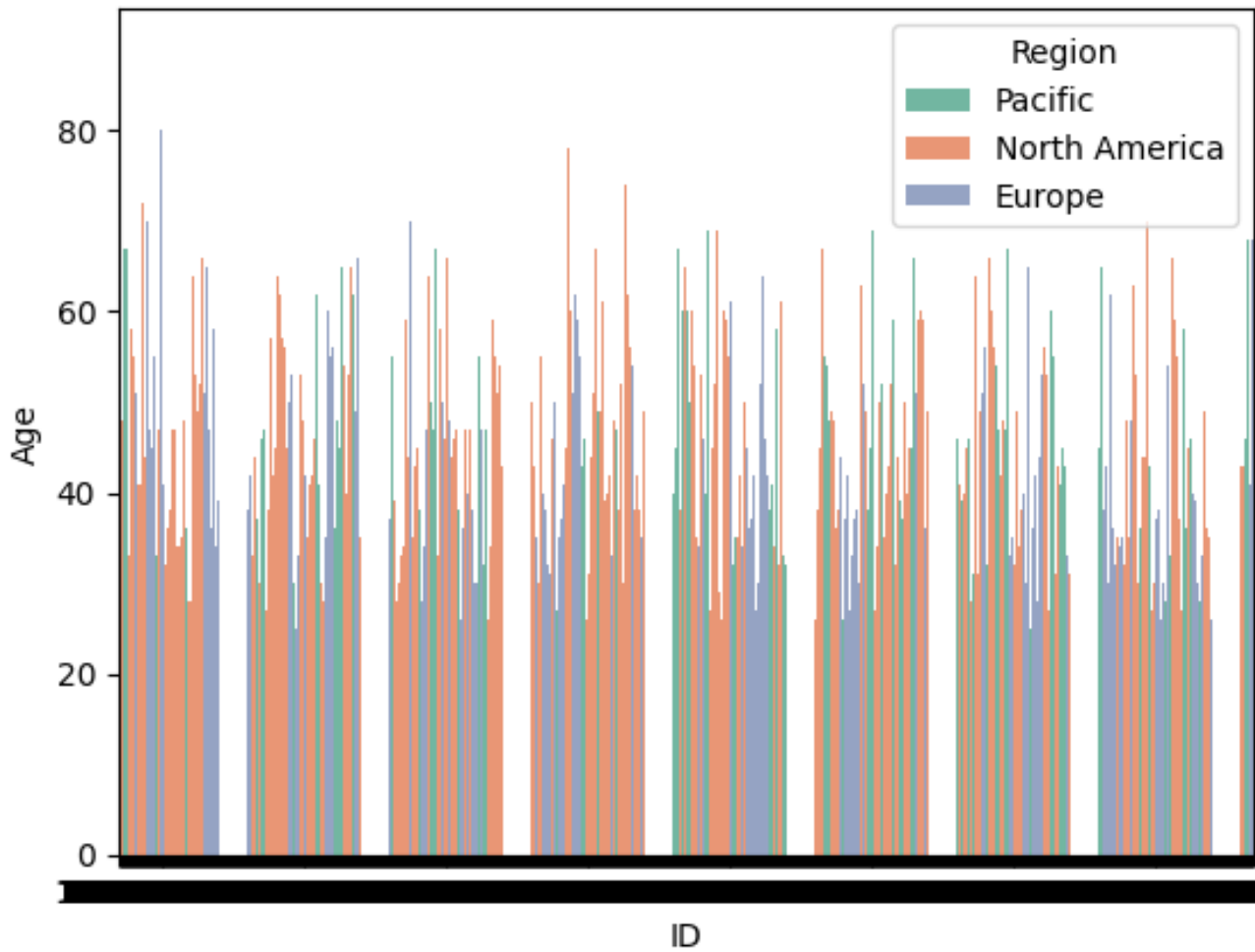
Code

```
#color change in chart
sns.barplot(x=data['ID'],y=data['Age'],hue=data['Region'],palette="Set2")
plt.xlabel("ID")
plt.ylabel("Age")
plt.title("Bar Chart")
plt.show()
```

Output

<Figure size 640x480 with 1 Axes>

Bar Chart

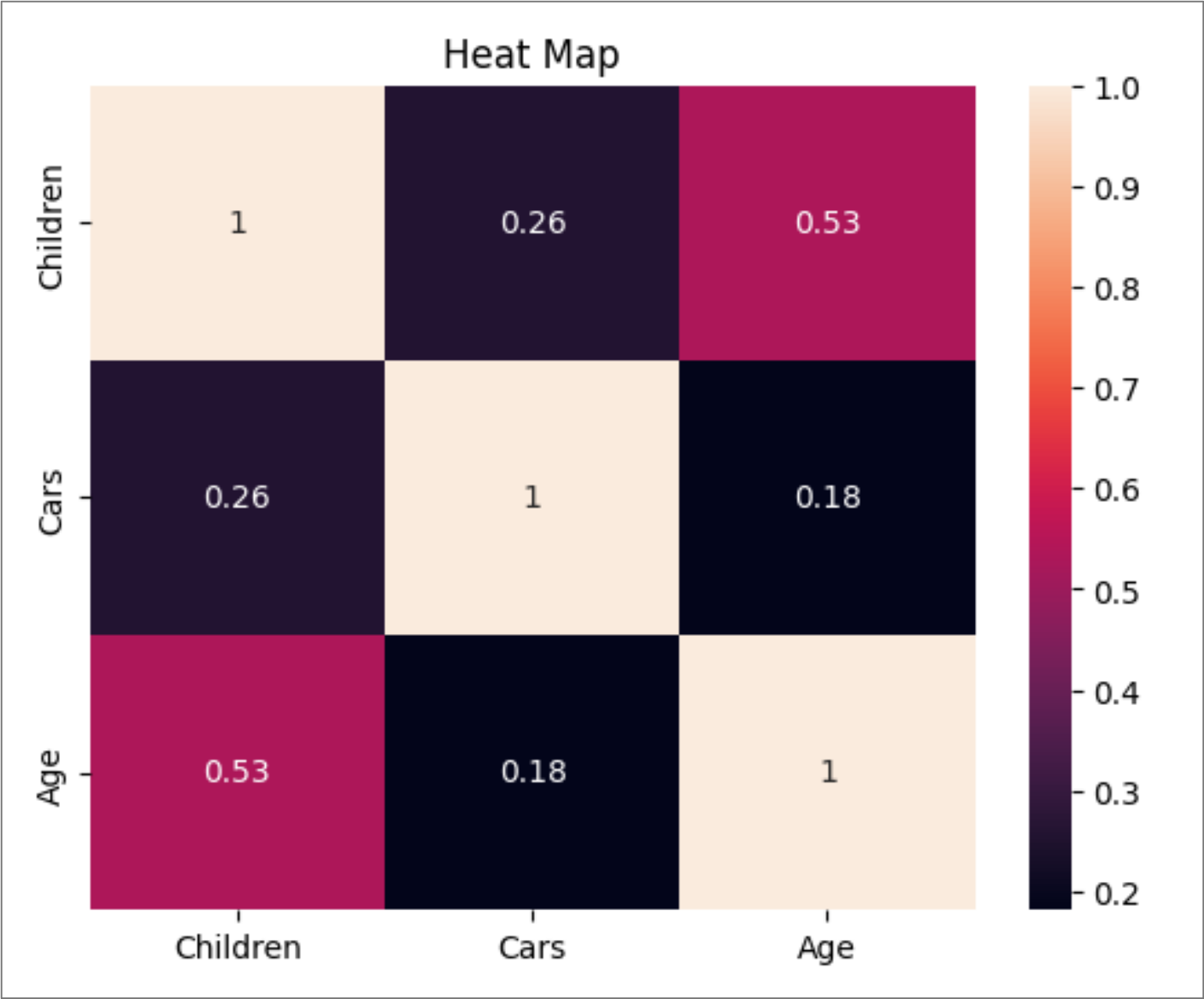


Code

```
#Heat map
sns.heatmap(d1.select_dtypes(include=[np.number]).corr(),annot=True)
plt.title("Heat Map")
plt.show()
```

Output

<Figure size 640x480 with 2 Axes>



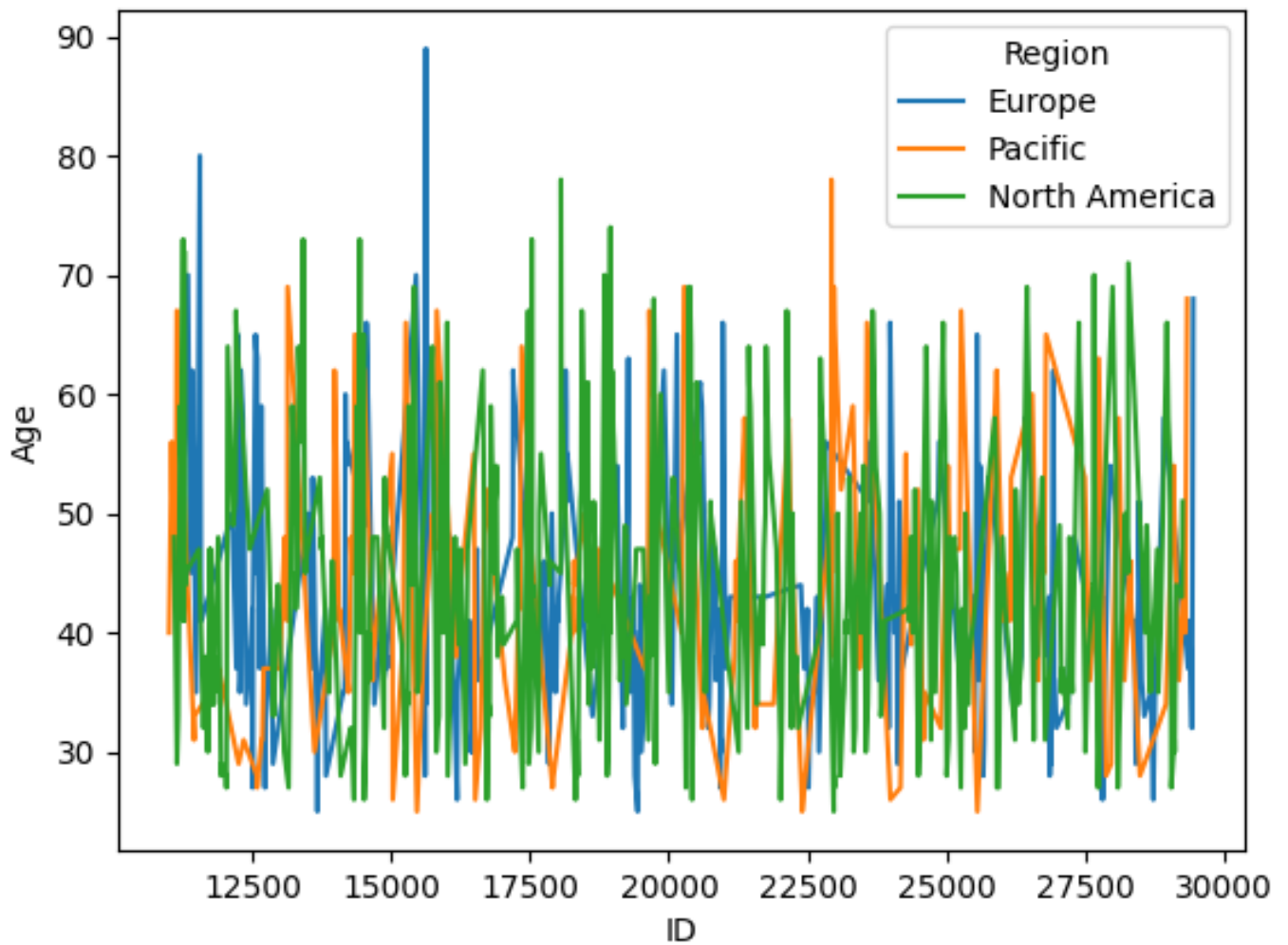
Code

```
#Line chart for multivariate
sns.lineplot(x=data['ID'],y=data['Age'],hue=data['Region'])
plt.xlabel("ID")
plt.ylabel("Age")
plt.title("Line Chart")
plt.show()
```

Output

<Figure size 640x480 with 1 Axes>

Line Chart



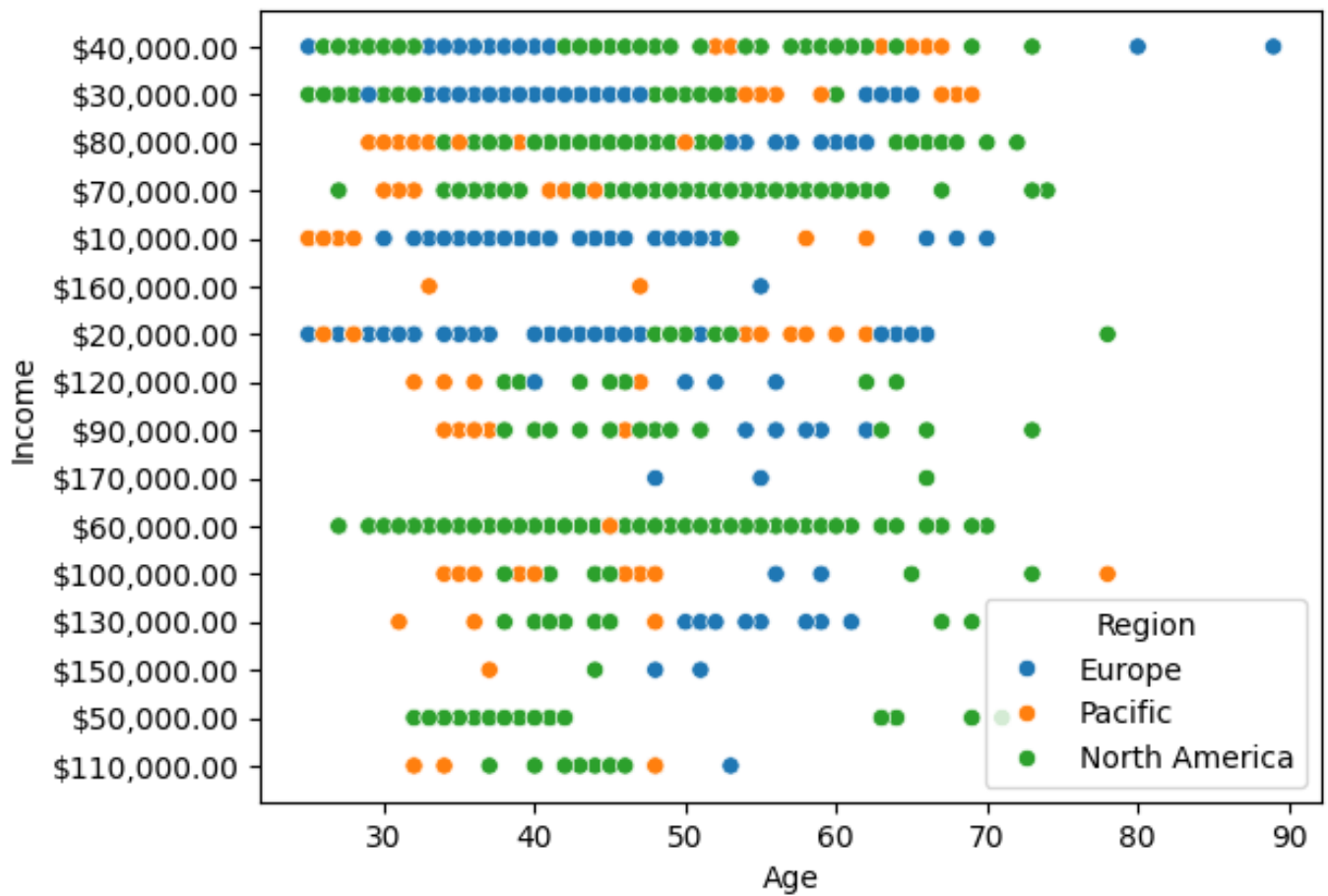
Code

```
#scatter plot
sns.scatterplot(x=data['Age'],y=data['Income'],hue=data['Region'])
plt.xlabel("Age")
plt.ylabel("Income")
plt.title("Scatter Plot")
plt.show()
```

Output

<Figure size 640x480 with 1 Axes>

# Scatter Plot

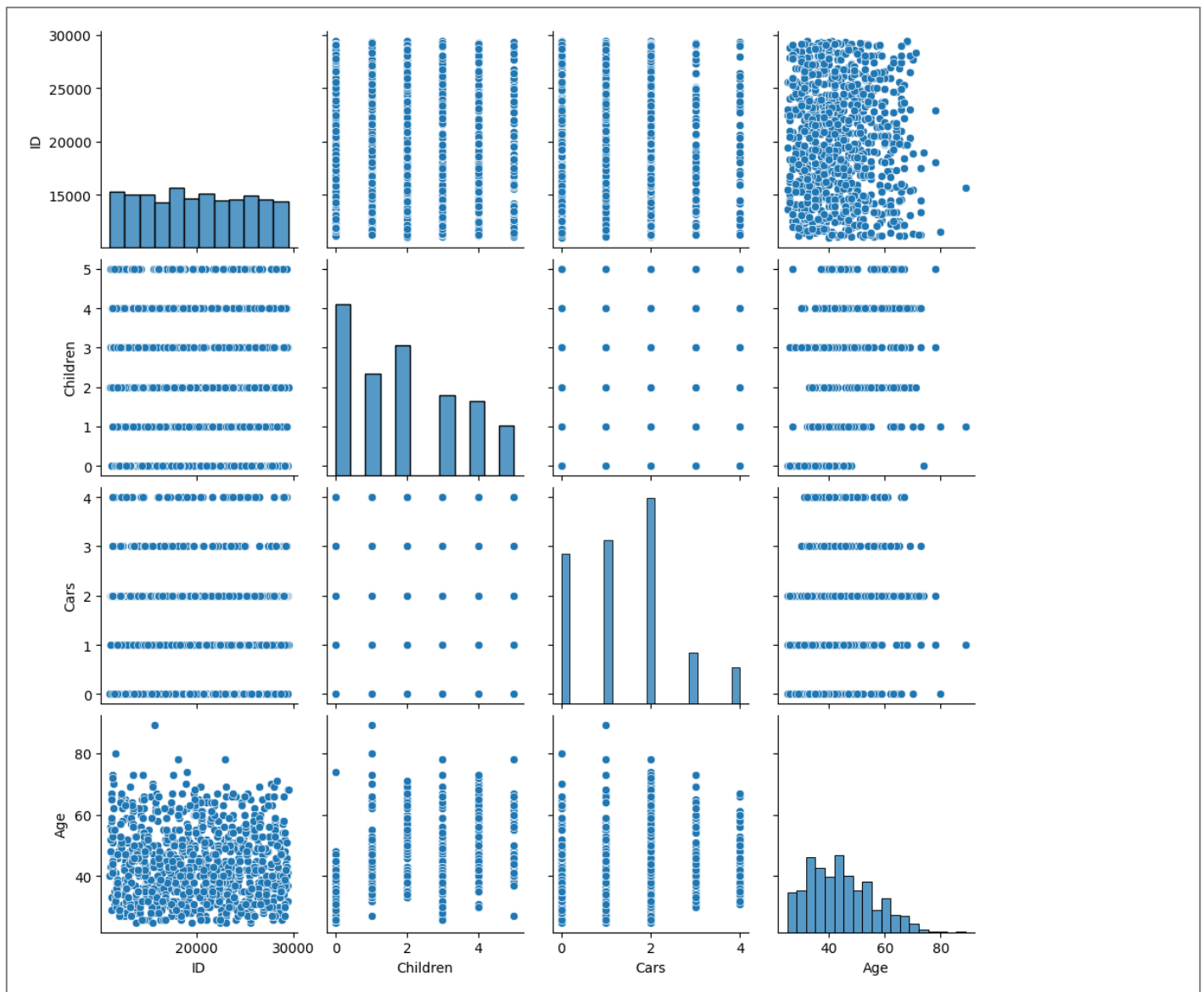


Code

```
#pairplot
sns.pairplot(data)
plt.show()
```

Output

<Figure size 1000x1000 with 20 Axes>



#### Code

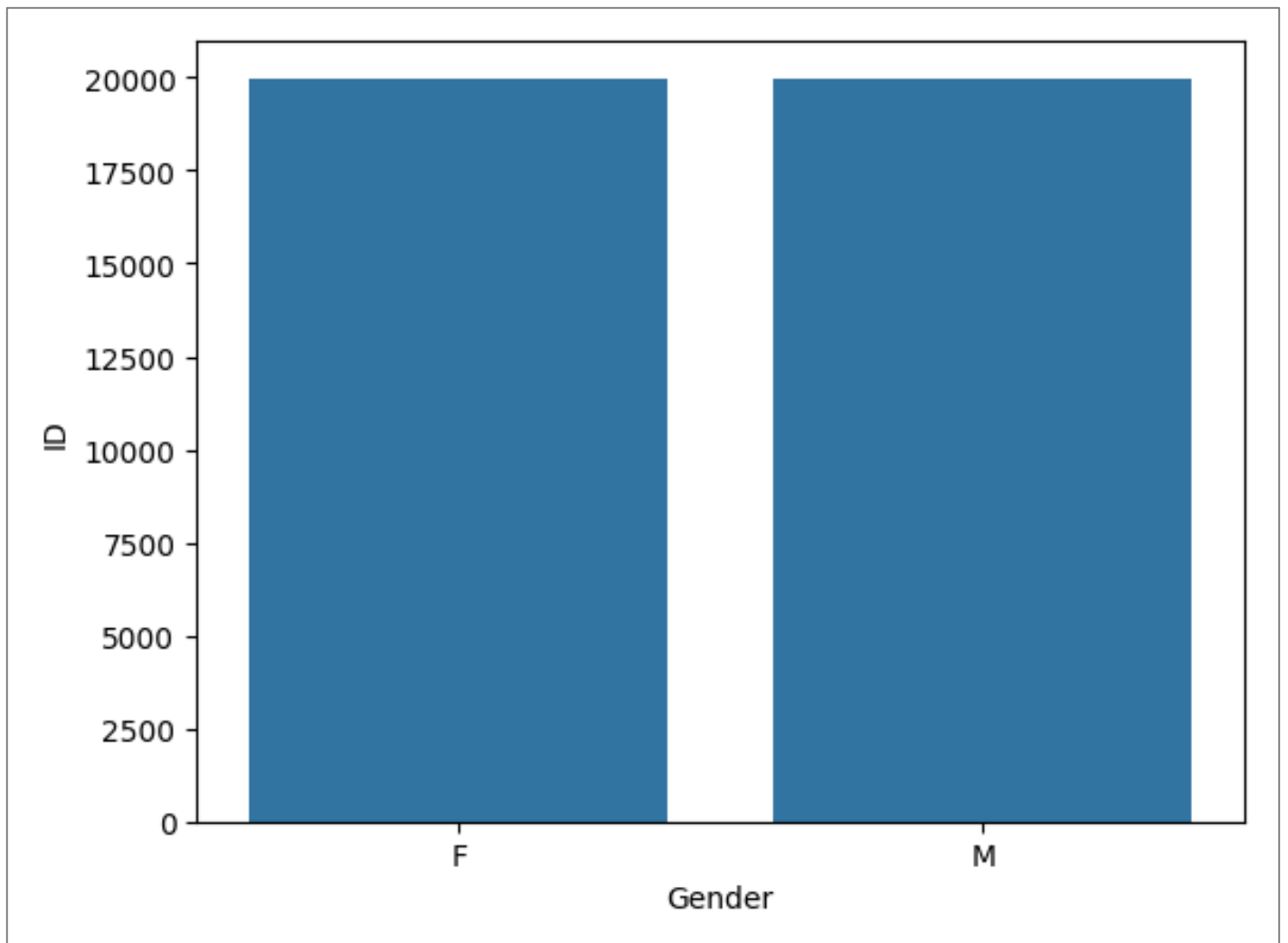
```
#statistical analysi chart-groupby
sns.barplot(x=data.groupby("Gender")["ID"].mean().index,y=data.groupby("Gender")["ID"].mean())
```

#### Output

```
<Axes: xlabel='Gender', ylabel='ID'>
```

#### Output

```
<Figure size 640x480 with 1 Axes>
```

**Code**

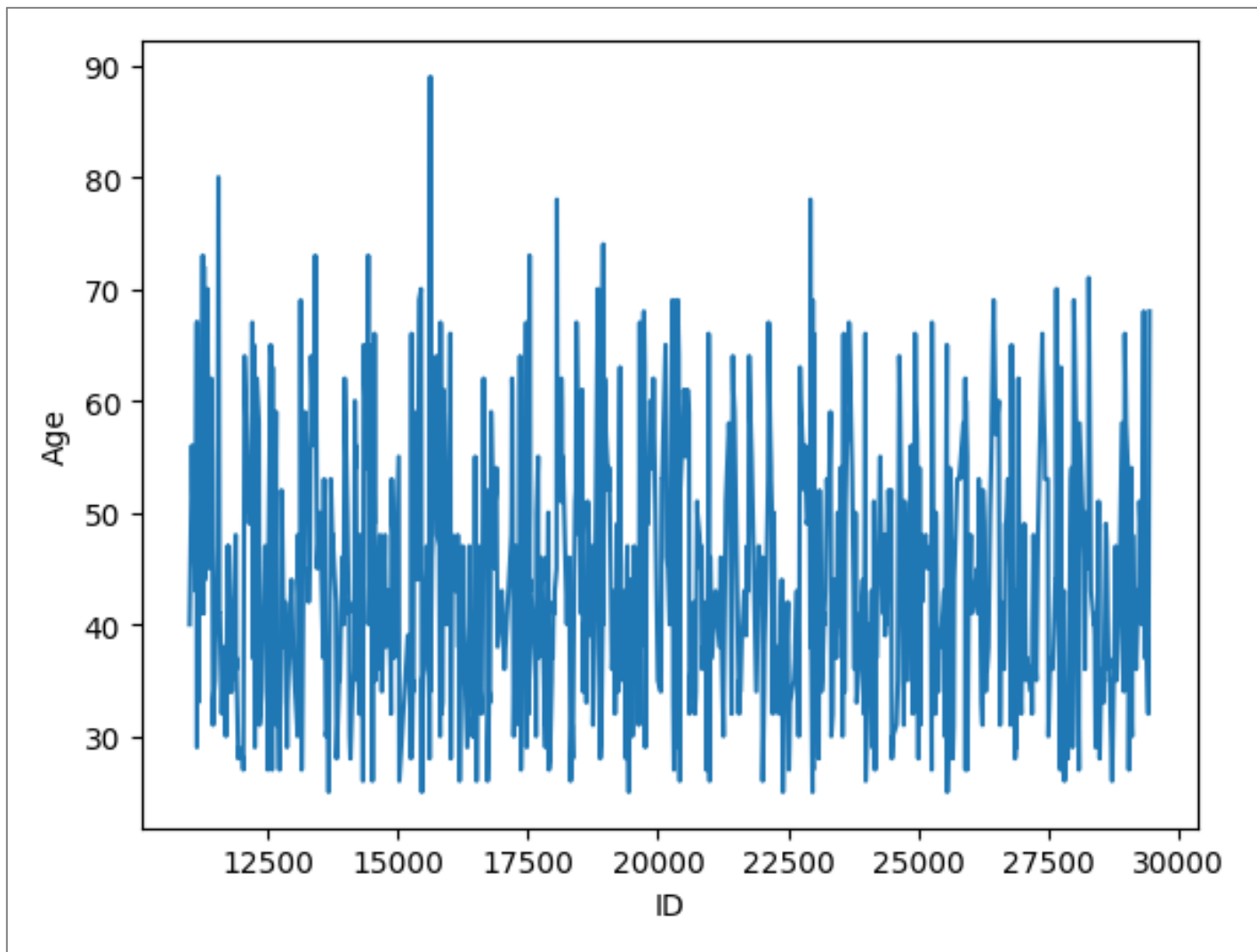
```
#maximum chart  
sns.lineplot(x=data.groupby("ID")["Age"].max().index,y=data.groupby("ID")["Age"].max())
```

**Output**

```
<Axes: xlabel='ID', ylabel='Age'>
```

**Output**

```
<Figure size 640x480 with 1 Axes>
```



#### Code

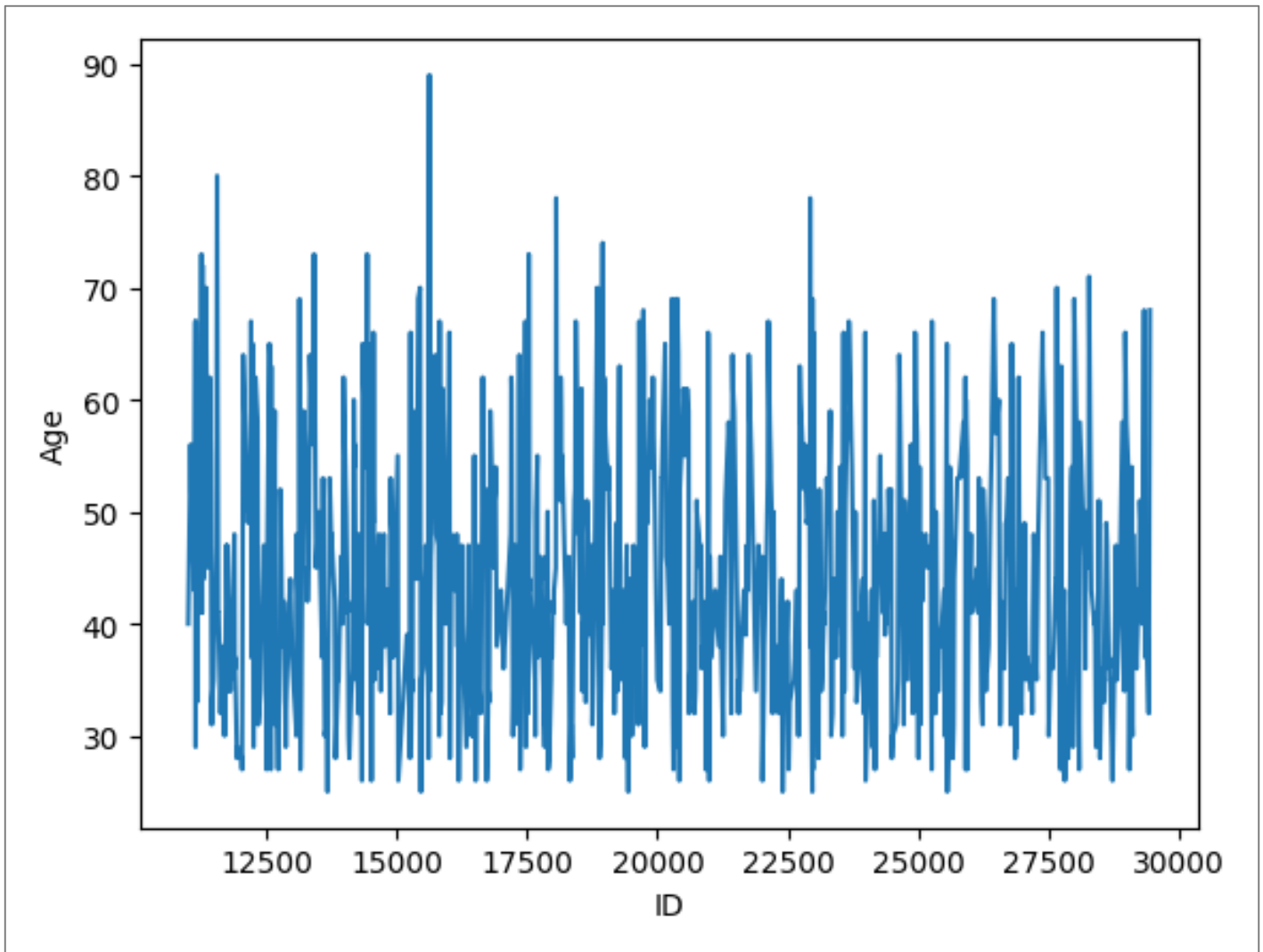
```
#maximum chart  
sns.lineplot(x=data.groupby("ID")["Age"].mean().index,y=data.groupby("ID")["Age"].mean())
```

#### Output

<Axes: xlabel='ID', ylabel='Age'>

#### Output

<Figure size 640x480 with 1 Axes>



**Descriptive Analysis:** The average age of customers in the dataset was approximately 44.16 years after the missing values were handled. The average number of children was approximately 1.90. The dataset includes customer information such as marital status, gender, income, education, occupation, home ownership, number of cars, commute distance, region, and bike purchase status.

**Insights from Age and Cars:** The analysis of average age based on the number of cars showed that customers owning two cars had the highest average age, approximately 47.79 years. Customers with zero cars had an average age of approximately 41.27 years. This indicates variations in the average customer age across different car ownership groups.

**Data Distribution:** The dataset contains customer ages ranging from 25 to 89 years. The median age was 43 years, while the standard deviation was approximately 11.35 years. This shows that the customers belong to a range of age groups.

**Bike Purchase Analysis:** The dataset includes the 'Purchased Bike' column, which can be used to analyze customer purchasing behavior. Customer characteristics such as income, age, commute distance, occupation, and number of cars can be compared with bike purchase status to identify patterns. The uploaded analysis contains the purchase status column, but it does not include calculated purchase percentages or group-wise purchase rates.

**Conclusion:** The Bike Sales dataset provides useful information about customer demographics, financial characteristics, and transportation-related attributes. Data cleaning, missing-value handling, feature engineering, and exploratory data analysis help prepare the dataset for further customer behavior analysis. Additional analysis of bike purchase status can provide insights into the factors associated with purchasing decisions.